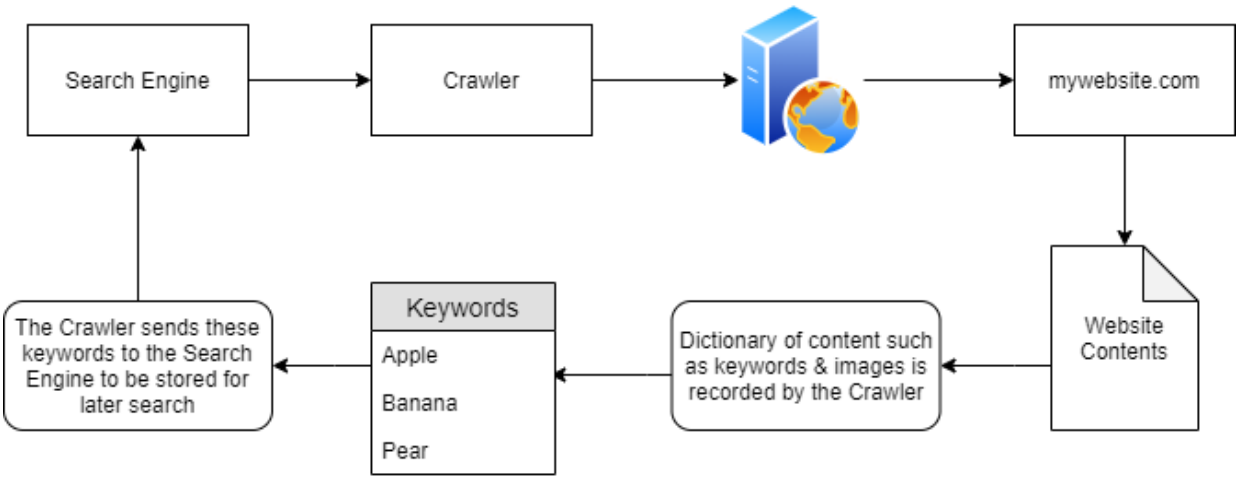
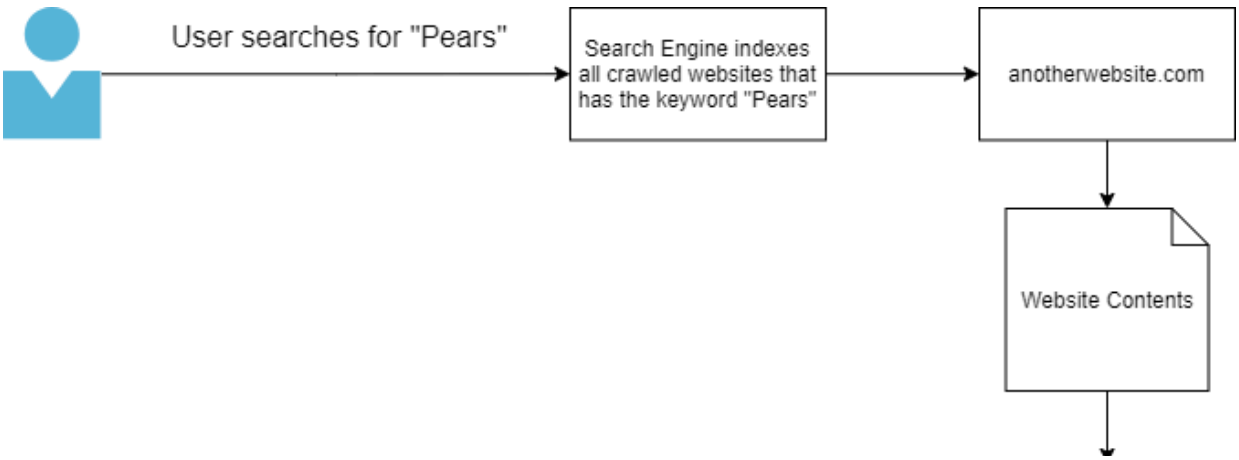
Room progress: 25%

The diagram below is a high-level abstraction of how these web crawlers work. Once a web crawler discovers a domain such as **mywebsite.com**, it will index the entire contents of the domain, looking for keywords and other miscellaneous information - but I will discuss this miscellaneous information later.



In the diagram above, "**mywebsite.com**" has been scraped as having the keywords as “Apple” “Banana" and “Pear”. These keywords are stored in a dictionary by the crawler, who then returns these to the search engine i.e. Google. Because of this persistence, Google now knows that the domain “**mywebsite.com**” has the keywords “Apple", “Banana” and “Pear”. As only one website has been crawled, if a user was to search for “Apple”...“**mywebsite.com**” would appear. This would result in the same behaviour if the user was to search for “Banana”. As the indexed contents from the crawler report the domain as having “Banana”, it will be displayed to the user.

As illustrated below, a user submits a query to the search engine of “Pears". Because the search engine only has the contents of one website that has been crawled with the keyword of “Pears” it will be the only domain that is presented to the user.